Survival-supervised deconvolution identifies a reproducible bulk tumor–stroma prognostic axis in pancreatic cancer

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Abstract

Pancreatic ductal adenocarcinoma (PDAC) outcomes reflect coordinated tumor-intrinsic and stromal states, but bulk transcriptomic methods typically decompose these compartments without survival supervision. We developed DeSurv, a survival-supervised matrix factorization framework that jointly learns gene programs and survival associations. In TCGA and CPTAC PDAC, DeSurv identified a compact three-program structure organized along a co-varying tumor–stroma axis (here, cross-sample covariation of tumor-cell and stromal gene programs rather than single-cell co-expression or spatial adjacency), with Classical tumor/restCAF-like stroma at one pole and Basal-like tumor/proCAF-like stroma at the high-risk pole. Across five external cohorts, each standard-deviation shift toward this pole was associated with a pooled hazard ratio of 1.50 (95% CI, 1.31–1.72) and remained prognostic after adjustment for PurIST and DeCAF; dichotomized high-risk patients had a hazard ratio of 1.79 (95% CI, 1.41–2.28). DeSurv's discrimination matched simpler survival-supervised predictors (penalized Cox, supervised principal components); its contribution is an interpretable, source-attributable decomposition of this signal rather than higher predictive accuracy. Projected without retraining into treated cohorts, the proCAF-like stromal program showed a consistently protective direction, reaching significance in the least-advanced cohort, consistent with therapy-associated remodeling. Together, these results identify a reproducible tumor–stroma prognostic axis in PDAC.

Pancreatic ductal adenocarcinoma (PDAC) is among the deadliest solid tumors, with a five-year relative survival rate of approximately 13%[1] and limited responsiveness to existing therapies. Bulk transcriptomic studies have defined reproducible malignant programs, most consistently Basal-like and Classical tumor states[2–4], as well as stromal programs including tumor-promoting proCAF and tumor-restraining restCAF states[5]. Single-cell and spatial studies further show that PDAC tumors comprise malignant, stromal, immune and other microenvironmental components whose transcriptional states can co-vary within the same lesion[6,7]. Whether joint configurations of these programs carry prognostic information beyond discrete tumor and stromal labels remains incompletely understood.

Bulk RNA profiling remains the primary platform for cohort-scale prognostic modeling[8,9], yet each profile is a composite of malignant, fibroblast, immune and other cellular sources[10]. Nonnegative matrix factorization (NMF) has become the dominant framework for resolving this mixture into additive, interpretable gene programs[3,11–13], yielding major PDAC insights including molecular subtypes[2,4,5,14], virtual microdissection of Basal-like and Classical programs[3], and compartment-specific deconvolution[13].

A limitation of this workflow is that programs are usually discovered without outcome information and tested for clinical association only after factorization[2,14]. Because unsupervised NMF minimizes reconstruction error, its factors tend to capture the highest-variance structure in the data, which may reflect tissue composition or other outcome-neutral variation rather than prognostic biology[8,15]. In PDAC, exocrine and other low-purity signals can dominate unsupervised factors while carrying limited prognostic value[4,5]. More broadly, tumor purity accounts for a large fraction of expression variation across cancer types[16], illustrating how reconstruction-optimal and outcome-relevant subspaces can diverge. Reconstruction alone also leaves the factor basis non-identifiable[17–19], contributing to cross-study variability among unsupervised cancer subtype taxonomies[20,21].

This raises a biological and methodological question: which bulk-transcriptomic programs carry reproducible prognostic information in PDAC, and how are those programs distributed across malignant, stromal, immune and other components of the tumor microenvironment? Neither extreme suffices on its own. Unsupervised factorization optimizes reconstruction and can over-capture dominant composition-driven variation, whereas selecting genes directly on survival can identify outcome-associated expression features but leaves their cellular source ambiguous. A gene's prognostic association may reflect malignant, stromal, immune or other microenvironmental variation.

Here we present DeSurv, a survival-supervised deconvolution framework that integrates NMF with Cox proportional hazards modeling so that gene programs are shaped during discovery by both reconstruction and survival. Supervision is applied to the gene-program matrix while preserving the mixture-coefficient interpretation of sample loadings, and trained programs score new samples by projection without retraining. We evaluate DeSurv through two non-circular tests: recovery of the predefined latent programs in simulations where the true gene sets are known a priori, and out-of-sample generalization to five independent PDAC cohorts on which no model parameters were estimated.

In simulations, DeSurv recovers prognostic programs more reliably than standard NMF, with advantages that scale with the separation between high-variance and prognostic subspaces and vanish when no survival signal is present. In PDAC, DeSurv recovers a compact three-program structure organized around a co-varying tumor–stroma axis, that is, covariation of tumor and stromal gene programs across patients rather than single-cell co-expression or spatial adjacency, linking Classical tumor identity and restCAF-like stroma at one pole with Basal-like tumor identity and proCAF-like stroma at the high-risk pole. This score generalizes across five independent PDAC cohorts without retraining and remains prognostic beyond established PurIST and DeCAF classifications. These findings identify a reproducible tumor–stroma prognostic axis in PDAC, recoverable de novo from bulk transcriptomes by survival-supervised matrix factorization.

Results

Model Overview

We developed DeSurv, a survival-supervised deconvolution framework that learns two linked quantities from bulk transcriptomes paired with patient outcomes: gene programs and their survival associations (Fig. 1A, Methods). DeSurv integrates nonnegative matrix factorization with Cox proportional hazards modeling so that each gene program is shaped during discovery by both expression reconstruction and survival association. The survival objective acts on the gene-program matrix rather than directly on sample loadings, preserving the mixture-coefficient interpretation of loadings and allowing trained programs to score new samples by projection without retraining. A supervision parameter controls the balance between reconstruction and survival, and this parameter, the factorization rank, and regularization are selected by Bayesian optimization on cross-validated concordance (Fig. 1B, Methods).

Applied to the PDAC training cohorts from TCGA[22] and CPTAC[23] (Materials and Methods), cross-validated concordance varied across candidate ranks but plateaued from $k = 3$ through $k = 12$ (SI Appendix, Fig. S4), indicating that three supervised factors captured nearly as much prognostic information as twelve. Within this plateau, joint Bayesian optimization over rank and supervision strength with a one-standard-error rule (Methods) selected the parsimonious model, $k = 3$ at $\alpha = 0.334$. The peak cross-validated C-index was 0.655 across 273 patients with 139 events. By contrast, standard NMF rank-selection diagnostics, including reconstruction residuals, cophenetic correlation and silhouette width[12,24], did not converge on a consistent rank in PDAC (SI Appendix, Fig. S5), motivating outcome-aware model selection. All rank and hyperparameter selection was performed within the TCGA and CPTAC training set, before any projection into validation cohorts. This $k = 3$ solution was dominated by a co-varying tumor–stroma axis, which we characterize next.

Survival supervision and standard NMF produce different factor structures in PDAC

To determine how survival supervision shaped the recovered factor structure, we fit standard NMF at the same rank selected for DeSurv ($k = 3$) and compared each method's factor-specific gene rankings with established PDAC gene programs (Fig. 2A–B). Matching k across methods isolates the effect of survival supervision on factor content from differences in model complexity; a sensitivity analysis using NMF's independently selected ranks (elbow $k = 5$, BO $k = 7$) is provided in SI Appendix, Table S5. We first asked which malignant, stromal, immune and other microenvironmental programs were represented in each factorization, then tested whether these structural differences corresponded to differences in survival contribution.

We refer to the three DeSurv factors as D1–D3 and the three NMF factors as N1–N3. In the DeSurv factorization, D1 correlated with Classical tumor programs and restCAF-like stromal signatures, D2 with proCAF-like programs, and D3 with Basal-like tumor programs (Fig. 2A). Thus, DeSurv organized the dominant survival-associated signal around a tumor–stroma axis: higher D1 scores mark a Classical/restCAF-like, lower-risk state, whereas movement toward Basal-like tumor and proCAF-like stroma marks higher risk. This organization is consistent with evidence that survival stratification in PDAC improves when tumor-intrinsic and CAF subtypes are considered jointly[5,25]. No DeSurv factor showed significant positive correlation with exocrine-associated programs, including

DECODER Exocrine, Collisson Exocrine and Bailey aberrantly differentiated endocrine–exocrine (ADEX) programs (Fig. 2A).

Standard NMF produced a different factor organization (Fig. 2B). One factor, N2, was strongly correlated with exocrine-associated programs and negatively correlated with immune and stromal signatures. A second factor, N1, correlated with malignant-epithelial identity and carried both Classical and Basal-like markers. A third factor, N3, correlated with immune, fibroblast and extracellular matrix (ECM)-related programs, including Puleo Desmoplastic, SCISSORS proCAF and restCAF[26], Maurer Immune and ECM[27], and DECODER Activated Stroma. Thus, at $k = 3$, standard NMF devoted one factor to exocrine-associated variation and aggregated tumor-intrinsic and microenvironmental programs into two broad factors, whereas DeSurv separated Basal-like tumor, proCAF-like stroma and a Classical/restCAF-like survival-aligned program.

NMF seeks a low-rank *reconstruction* of the transcriptome; many factorizations can reconstruct comparably well, and the unsupervised objective has no incentive to prefer prognostically informative directions. We therefore quantified each factor's contribution using two metrics (Fig. 2C; SI Appendix, Sections 10–11): a normalized reconstruction Shapley value, which partitions reconstruction contribution across factors and sums to 100%, and survival contribution, defined as the change in Cox partial log-likelihood ($\Delta\ell$) when that factor is dropped from a k-factor Cox model. NMF distributed reconstruction across factors in proportions unrelated to prognosis: all three NMF factors had negligible $\Delta\ell$ despite spanning a wide range of reconstruction contribution. By contrast, DeSurv concentrated nearly all survival contribution into D1. Specifically, D1 contributed 18.8% of reconstruction but $\Delta\ell = 66.4$, whereas the largest-reconstruction NMF factor, N1, contributed 41.8% of reconstruction but only $\Delta\ell = 0.1$ to survival. The exocrine-associated NMF factor, N2, contributed 18.6% of reconstruction but only $\Delta\ell = 1.16$ to survival. The largest-reconstruction DeSurv factor, D3, contributed 56% of reconstruction but added little survival information after accounting for D1. Thus, in PDAC, factors that best reconstruct expression are not necessarily the factors that best explain survival, consistent with tumor-purity analyses across cancer types[16].

To further characterize the relationship between the two factorizations, we examined pairwise W-matrix gene-loading correlations across all 1,970 genes (Fig. 2D; Spearman correlations over full loading profiles, since top-gene subsets can overstate preservation). At $k = 3$, NMF captured the malignant-epithelial compartment largely in a single factor, N1, which carried both Classical and Basal-like markers and therefore is labeled "tumor'' rather than Classical. N1 mapped strongly onto DeSurv's Basal-like tumor factor, D3 (Spearman $\rho = 0.84$), while NMF's microenvironmental factor, N3, mapped onto DeSurv's proCAF-like stromal factor, D2 ($\rho = 0.87$). In contrast, DeSurv's most prognostic factor, D1, was only weakly correlated with all three NMF factors (all $\rho \leq 0.36$). Because D1 couples Classical tumor and restCAF-like stromal signatures into a single survival-aligned program, it is not simply a reweighting of an NMF factor (SI Appendix, Table S2). NMF's exocrine factor, N2, had no strong DeSurv counterpart (maximum $\rho = 0.61$), consistent with reduced emphasis on exocrine-associated variation under survival supervision. Together, these analyses show that standard NMF primarily separates broad tumor, stromal/immune and exocrine-associated structure, whereas DeSurv resolves a survival-aligned tumor–stroma program that standard NMF does not isolate.

Survival-aligned programs generalize across independent PDAC cohorts with lower complexity

A prognostic factorization is useful only if it generalizes beyond the training cohort. Having established that survival supervision and standard NMF produce different factor structures in the training data, we next tested whether each method's learned programs transferred to independent cohorts without retraining. The five external validation cohorts (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq and Puleo) consisted of non-metastatic, treatment-naive specimens (SI Appendix, Table S1). We computed factor scores in each validation dataset by projecting the gene programs learned in the training cohort onto each new expression matrix, $Z = \widetilde{W}^\top X_{\text{new}}$, yielding a continuous score for each factor in each sample.

External validation C-indices for NMF at independently selected ranks are summarized in SI Appendix, Table S5. Standard NMF at elbow-selected $k = 5$ and Bayesian-optimization-selected $k = 7$ substantially improved over NMF at $k = 3$, with $k = 7$ matching or exceeding DeSurv's per-cohort C-index in most cohorts. DeSurv's aim, however, is not maximal prediction at arbitrary rank but recovery of a compact, interpretable tumor–stroma program; the NMF result required more than twice as many factors and depended on applying DeSurv's tuning framework with $\alpha = 0$ in the joint optimization over rank and hyperparameters, whereas NMF's own rank-selection diagnostics (cophenetic correlation, silhouette width and reconstruction residuals) did not converge on this rank (SI Appendix, Fig. S5). DeSurv at $k = 3$ retained a significant association with survival after simultaneous adjustment for PurIST and DeCAF classifications (per-SD HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$; Table 1), indicating independent prognostic value as a lower-dimensional, biologically coherent program. Standard NMF at $k = 7$ attained an adjusted hazard ratio at least as large (Table 1); we therefore do not claim predictive superiority for DeSurv, whose advantage is recovering this association as a compact, interpretable, source-attributable program rather than as a higher-dimensional, less-interpretable factorization.

Prediction alone did not distinguish DeSurv from simpler survival-supervised models. Freezing each method's leading prognostic direction on the training data and projecting it without refitting onto the five validation cohorts, DeSurv's pooled transfer concordance (C-index = 0.62) was on par with survival-supervised baselines fit directly on genes: penalized (lasso) Cox (0.61), Cox partial least squares (0.63) and supervised principal components (0.64). Unsupervised directions transferred poorly (standard NMF at $\alpha = 0$, 0.55; the leading principal component, 0.49), consistent with evidence that learned representations do not reliably out-predict penalized regression on genes for out-of-sample prognosis[28]. Because these survival-supervised scores are nearly interchangeable for prediction yet collapse the signal into a single, compartment-ambiguous axis, DeSurv's contribution is not higher accuracy but the decomposition of that same prognostic signal into interpretable, source-attributable programs.

When β coefficients were re-fit within each validation cohort, DeSurv's three-factor decomposition was preferred by BIC over NMF's seven-factor decomposition in three of five cohorts, with the remaining two essentially tied (SI Appendix, Table S6). Across the 5 independent PDAC cohorts ($n = 616$), the DeSurv linear predictor (oriented so that higher values indicate higher risk) showed a consistent survival association in a stratified Cox model (pooled HR per SD = 1.50, 95% CI 1.31–1.72, $P = 2.6 \times 10^{-9}$; per-cohort forest plot in Fig. 3A; Kaplan–Meier curves in SI Appendix, Fig. S6C–F). At the matched rank of $k = 3$, the analogous NMF linear predictor showed a weaker pooled association (HR per SD = 1.11, 95% CI 1.00–1.23, $P = 0.057$), motivating a matched-rank comparison of dichotomized risk groups.

The continuous, stratified-Cox associations above are our primary analysis; to illustrate them clinically, we dichotomized validation samples into high- and low-risk groups using method-specific cutpoints independently optimized by cross-validation in the training data. The DeSurv linear predictor separated survival trajectories in the pooled validation cohort (HR = 1.79, 95% CI 1.41–2.28, $P < 0.001$; Fig. 3B). Applying the same procedure to NMF at $k = 3$ yielded weaker stratification (HR = 1.38, 95% CI 0.92–2.09, $P = 0.122$; Fig. 3C). The DeSurv cutpoint classified a larger validation subset as high risk than NMF at matched rank (151 versus 37 high-risk patients; 114 reclassified) while preserving strong survival separation. The DeSurv high-risk group was enriched for both established poor-prognosis subtypes, Basal-like by PurIST and proCAF by DeCAF (Fisher's exact test; SI Appendix, Fig. S7), capturing consensus biology while adding independent prognostic information. By contrast, NMF risk groups at $k = 3$ showed Basal-like enrichment but no significant proCAF enrichment (SI Appendix, Fig. S7), consistent with NMF not isolating the tumor–stroma axis represented by D1.

Together, these results indicate that survival supervision yields a compact, biologically interpretable prognostic structure that generalizes across independent PDAC cohorts. Higher-rank NMF models can achieve similar per-cohort concordance, but only at greater complexity and without recovering the same axis; DeSurv instead identifies a lower-dimensional tumor–stroma program with prognostic value beyond PurIST and DeCAF.

Table 1: Pooled external validation hazard ratios (per SD of the linear predictor) for DeSurv at $k = 3$ and standard NMF at BO-selected $k = 7$ ($\alpha = 0$), across the five validation cohorts ($n = 616$ patients, 414 events). Adjusted models include the PurIST tumor-intrinsic and DeCAF stromal molecular classifiers as covariates, with stratification by dataset. Both methods retain significant prognostic value after adjustment, with DeSurv's larger HR drop reflecting biological concordance with established subtypes and NMF $k = 7$'s smaller drop reflecting orthogonal residual content from its additional, less interpretable factors.

Method	Unadjusted HR per SD (95% CI), P	Adjusted HR per SD (95% CI), P
DeSurv ($k = 3$)	1.50 (1.31–1.72), < 0.001	1.33 (1.13–1.56), < 0.001
Std NMF ($k = 7$, BO $\alpha = 0$)	1.48 (1.34–1.64), < 0.001	1.47 (1.27–1.70), < 0.001

The tumor–stroma axis adds prognostic information beyond established PDAC classifiers

To establish that the dominant program is more than a continuous restatement of discrete subtype calls, we partitioned D1 against both classifiers across the pooled validation cohorts ($n = 616$, 414 events). Binary PurIST and DeCAF calls together accounted for 28% of the within-cohort variance of the continuous D1 score, indicating that D1 grades tumor–stroma state more finely than the dichotomous subtypes with which it aligns. More importantly, D1 retained significant prognostic value after conditioning on both classifiers (adjusted HR per SD = 0.82, 95% CI 0.74–0.92, $P = 0.0006$; partial likelihood-ratio $\chi^2 = 11.6$ on 1 df, $P = 0.0006$), improving stratified-Cox concordance over a PurIST-plus-DeCAF model from 0.593 to 0.621. The tumor–stroma axis therefore carries prognostic information that neither the tumor-intrinsic nor stromal classifier captures alone. This axis was also not specific to DeSurv's optimizer: three independent survival-supervised methods (supervised principal components, Cox partial least squares and sparse Cox) recovered the same D1 score in the training cohort (mean $|r| = 0.83$), whereas the leading unsupervised direction did not ($|r| = 0.10$), identifying D1 as a property of survival supervision rather than of a single algorithm (SI Appendix, supervised-method recovery). After adjustment for standard clinical prognostic variables (stage, grade, age and sex), the DeSurv score retained a significant, independent association with survival in the fully-annotated external PACA-AU cohorts, and it remained prognostic after adjustment for tumor purity, arguing against a purity- or composition-driven artifact (SI Appendix, Table S4); the smaller Dijk and Moffitt cohorts were individually underpowered for multivariable analysis.

The recovered programs show context-dependent prognostic associations in treated PDAC

As an exploratory, hypothesis-generating analysis, we examined how the fixed, treatment-naive programs behave when read in treated cohorts; because treatment, stage and cohort are confounded in these data, the associations below are prognostic, not predictive. Because DeSurv scores new samples by projecting the trained gene-program matrix without re-estimating parameters, the same programs can be read directly in these cohorts and compared across disease settings. We projected the trained programs onto three bulk treated cohorts: the metastatic gemcitabine-backbone trials Rash ($n = 85$) and Accept ($n = 70$), and the borderline-resectable/locally advanced Linehan cohort (FOLFIRINOX $\pm$ CCR2 inhibitor, $n = 66$). Because these cohorts differ by stage and regimen, we analyzed each separately rather than as a pooled estimate (Fig. 4; SI Appendix, treated-cohort analysis). Because the elastic-net penalty shrank the trained survival coefficient for the proCAF program D2 to exactly zero in the treatment-naive model (an estimated selection outcome, not an imposed constraint), D2 contributes nothing to the trained risk score, and programs were evaluated per factor; any association of D2 in the treated cohorts therefore reflects a marginal re-evaluation of the projected program in a new clinical context rather than a property of the trained predictor.

The proCAF stromal program D2 was associated with longer overall survival in all three treated cohorts: Linehan, adjusted HR per SD = 0.46 (95% CI 0.31–0.70, $P < 0.001$); Accept, 0.73 (95% CI 0.52–1.04, $P = 0.078$); and Rash, 0.77 (95% CI 0.53–1.12, $P = 0.178$). Thus, D2 showed a consistently protective direction across independent settings and chemotherapy backbones, reaching significance in the borderline-resectable/locally advanced cohort and remaining directionally concordant in the metastatic cohorts. This association was not attenuated by adjustment for the tumor-intrinsic and CAF classifiers; in Linehan, where D2 reached significance, the adjusted effect strengthened rather than weakened. D2 was also associated with longer progression-free survival in the metastatic setting and increased from pre- to post-treatment in paired biopsies, consistent with therapy-associated remodeling of the proCAF program (SI Appendix). Near-identical results under full-gene projection (Spearman $r \geq 0.86$ between top-gene and full-gene scores; SI Appendix) further supported the robustness of this association.

By contrast, D1 showed a setting-dependent association that tracked the basal–classical axis rather than a property unique to D1. D1 separated PurIST calls in every treated cohort (AUC 0.84–0.89), remained prognostic in the less-advanced Linehan cohort with a marginal HR per SD of 0.58 (95% CI 0.36–0.94, $P = 0.027$), and attenuated in the metastatic cohorts. Established basal–classical classifiers showed the same pattern: PurIST, Moffitt and Bailey classifications were all non-significant in the metastatic cohorts (SI Appendix). These findings indicate stage-dependent attenuation of the basal–classical prognostic axis rather than a failure specific to D1. Basal-like prevalence in the metastatic cohorts was within the expected range (16–22%), and the same axis was strongly prognostic in treatment-naive cohorts with similar prevalence ($P < 10^{-8}$); the Linehan estimate was based on only six Basal-like cases and was correspondingly imprecise, and large real-world cohorts confirm that the basal subtype remains adverse in metastatic disease[29]. We therefore report D1 marginally, because adjustment for the collinear PurIST classifier would condition the same axis on itself.

D3, the Basal-like program, was not associated with overall survival in any treated cohort, consistent with the basal–classical contrast residing primarily in D1. However, D3 was associated with shorter progression-free survival in the metastatic setting, with an HR per SD of 1.35 (95% CI 1.08–1.69, $P = 0.009$), consistent with the aggressive biology of Basal-like tumors. The D1 score also independently validated against GATA6 RNA in situ hybridization (Spearman $\rho = 0.68$, $P < 0.001$; SI Appendix). Because treatment was confounded with cohort and stage, these analyses address prognosis rather than treatment prediction.

DeSurv recovers prognostic gene programs when variance and prognosis diverge

We asked whether survival supervision improves recovery of prognostic gene programs when prognostic variation is not the dominant source of transcriptomic variance. To address this, we designed simulations with known ground-truth latent structure, in which prognostic gene programs explained low variance relative to outcome-neutral background signals and survival times were generated from latent factor scores via a Cox model ($p = 3{,}000$ genes, $n = 200$ samples, true $k = 3$; Materials and Methods, Simulation Studies). These simulations probe this specific regime, in which survival-relevant programs are weaker than variance-dominant background structure, rather than representing all transcriptomic settings. To test whether incorporating survival information during factorization improves recovery, rather than estimating it only afterward, we compared DeSurv to standard NMF ($\alpha = 0$), where factors are learned by reconstruction error alone and survival associations are estimated by a separately fit Cox model. Both methods were tuned via BO over cross-validated C-index, so that differences in performance reflect the role of supervision during factorization, not during model selection.

In the prognostic scenario, where prognostic programs explained low variance relative to outcome-neutral programs, DeSurv consistently achieved higher C-index (Fig. 5A) and substantially higher precision for recovering the true prognostic gene programs (Fig. 5B). Here precision measures how well the learned factors recover the true prognostic programs, computed per program against its best-matching factor and averaged (SI Appendix, Sections 6 and 9). The unsupervised baseline exhibited near-zero precision, indicating that variance-driven factorization failed to concentrate on the genes that actually drove survival. This indicates that outcome supervision during factorization can be required to recover prognostic programs that do not dominate expression variance. Across 100 replicates, DeSurv achieved median C-index 0.837 versus 0.724 for standard NMF ($\Delta = 0.113$, paired Wilcoxon $P < 0.001$). The gain in gene-program precision exceeded the gain in C-index, indicating that unsupervised factorization can achieve similar predictive performance while recovering different gene programs. Survival supervision therefore improves not only prediction but also recovery of the underlying prognostic biology. A similar decoupling appears empirically in PDAC at $k = 5$, where supervised (DeSurv) and unsupervised (NMF) factorizations yield substantially different factor-specific top genes (SI Appendix, Fig. S8 [DeSurv $k = 5$] vs Fig. S9 [NMF $k = 5$]), with similar C-indices in external cohorts.

To verify that these gains reflect genuine signal recovery rather than overfitting, we repeated the analysis under a null scenario ($\beta = 0$) and a mixed scenario with partial overlap between variance-dominant and prognostic structure (SI Appendix, Fig. S3). The benefit of survival supervision scaled with the separation between prognostic and variance-dominant structure: largest in the prognostic scenario ($\Delta = 0.113$), attenuated when variance and prognosis partially overlapped ($\Delta = 0.069$, paired Wilcoxon $P < 0.001$), and absent when no survival signal existed, where both methods yielded C-index near 0.5. DeSurv also recovered the true rank ($k = 3$) more frequently than standard NMF, which systematically under-selected near $k = 2$ (Fig. 5C).

Discussion

Survival-supervised factorization shows that PDAC's prognostic transcriptional structure is dominated by a single axis linking Classical tumor identity with restCAF stroma, rather than by tumor-intrinsic and stromal programs considered separately. This axis generalizes across five independent PDAC cohorts without retraining and remains significant after adjustment for PurIST and DeCAF, indicating that PDAC outcome is encoded by the coordinated configuration of tumor-intrinsic lineage state and stromal composition, rather than by either compartment alone. A purely supervised predictor fit on bulk expression can match this external prediction, but returns a single, compartment-ambiguous risk score rather than source-resolved programs. DeSurv instead concentrates prognostic signal into fewer, interpretable gene programs by coupling NMF to a Cox objective during discovery: supervision acts on the gene-program matrix (W) while preserving the mixture interpretation of sample loadings (H), enabling single-sample scoring by projection, a transportability property not shared by methods that route supervision through H[30,31]. More broadly, prior survival-supervised representation methods, including interpretable autoencoder–Cox models, learn latent features whose prognostic signal is entangled with sample-specific loadings[32,33]; DeSurv's distinction is to keep supervision on the transferable gene-program basis W, so the learned programs remain source-attributable and transportable without re-estimation.

Methodologically, DeSurv recovers these programs de novo, without predefined signatures, while directly estimating their survival associations during factorization. Biologically, it isolates the tumor–stroma axis as a single survival-aligned program: the factor with the largest survival contribution, D1, spans Classical-tumor and restCAF-stromal genes, concordant with virtual[3] and experimental[27] microdissection, unsupervised deconvolution[13], and evidence that outcome depends on the joint configuration of tumor-intrinsic and fibroblast subtypes[5,25,34–36]. Standard NMF, by contrast, spent a dedicated factor on exocrine-compositional variation that reflects tumor purity rather than cancer biology[4], leaving less capacity for the microenvironmental programs DeSurv separates at the matched rank of $k = 3$.

This parsimony follows from the model-selection framework as much as from the survival gradient. Survival supervision creates a broad concordance plateau ($k = 3$–12; SI Appendix, Fig. S4) over which the one-standard-error rule selects a compact rank; standard NMF approached DeSurv's concordance only near $k = 7$ (SI Appendix, Table S5), with more than twice as many factors, and its own unsupervised rank-selection heuristics did not point there (SI Appendix, Fig. S5). At $k = 3$, DeSurv nonetheless retains independent prognostic value after classifier adjustment (Table 1). Joint Bayesian optimization over rank and supervision strength with the one-standard-error rule is therefore part of the methodological contribution: parsimony is a property of the rule, but supervision creates the flat surface on which the rule can act.

Our simulations clarify when this tradeoff is most beneficial: DeSurv's advantage is greatest when prognostic programs explain modest variance relative to outcome-neutral signals and vanishes under null conditions where no survival signal exists. DeSurv is therefore not intended to replace unsupervised NMF in all settings. For exploratory discovery without clinical endpoints, when survival data are sparse or unreliable, or when comprehensive biological characterization is the goal, unsupervised methods remain appropriate[37]. The supervision weight, tuned by Bayesian optimization, lets the data set the balance for a given cancer type and cohort size, with the $\alpha = 0$ endpoint recovering standard NMF as a special case. Nor is DeSurv proposed as a competing molecular classifier; the relevant question is whether its factors add prognostic signal beyond existing classifiers, which Table 1 confirms (adjusted HR = 1.33, 95% CI 1.13–1.56, $P < 0.001$). DeSurv is likewise distinct from reference-based deconvolution: methods such as CIBERSORTx[38] and BayesPrism[39] use single-cell or sorted references to estimate the cellular composition of a bulk sample, answering which cell types are present, whereas DeSurv learns prognostic latent gene programs directly from bulk expression without a cellular reference, asking which transcriptional programs carry outcome information and how they distribute across tumor and stromal compartments. The two are complementary, and DeSurv's source-attributable programs could in principle be intersected with deconvolution-derived cell fractions to test compartmental attribution more finely.

Several limitations define the scope of these findings. DeSurv assumes Cox proportional hazards, which may not hold under delayed or non-proportional effects, and the convergence guarantee (SI Appendix, Theorem 1) applies to an idealized algorithm; extensions to more flexible survival models are a natural direction. The method was applied here only to PDAC; the general methodological claim, that survival supervision recovers prognostic programs wherever the highest-variance and most-prognostic subspaces diverge, is therefore demonstrated for PDAC but remains to be established across cancer types, which will require multi-cancer benchmarking, including against strong penalized-regression baselines that learned representations do not consistently outperform for out-of-sample prognosis[28]. The training and primary validation cohorts are non-metastatic and treatment-naive, so the model is learned in resectable disease; projecting the trained programs into small treated cohorts (n = 66–85) suggested context-dependent remodeling of the proCAF and basal–classical axes, consistent with growing evidence that PDAC stroma is functionally dual[40–42] (Results; SI Appendix). Because treatment, stage and cohort are confounded and these cohorts are small, we regard these as exploratory, hypothesis-generating observations; whether the proCAF program identifies patients who benefit from stroma- or myeloid-directed therapy will require prospective trials. Finally, this axis is defined at the level of bulk gene programs, namely the covariation of Classical-tumor and restCAF-stromal signatures across patients, not per-cell co-expression or spatial adjacency, which single-cell and spatial assays are better positioned to test. That this covariation is not an artifact of tumor purity is supported by D1 remaining prognostic after PurIST/DeCAF adjustment and by its recovery by independent survival-supervised methods but not by the leading unsupervised direction. D1's Classical-tumor identity is further corroborated by GATA6 RNA in situ hybridization (Results). Adequately powered single-cell and spatial profiling of these cohorts is the priority orthogonal validation; resolving this axis at cellular resolution is nonetheless a future direction rather than a validation on which this bulk finding depends, and mechanistic experiments will be needed to determine whether these tumor–stroma configurations causally shape PDAC progression.

More broadly, the conventional workflow of unsupervised discovery followed by retrospective evaluation incurs a structural cost wherever the highest-variance signals are not the most prognostic, likely the norm where tumor purity and tissue composition dominate expression variation[16]. Aligning factorization with the clinical question from the outset avoids that cost: in PDAC, it recovered a Classical-tumor + restCAF program at $k = 3$, and the resulting DeSurv predictor transferred to five cohorts (pooled HR per SD 1.50) and survived PurIST/DeCAF adjustment. Supervision also stabilizes which genes define each program, a basis that reconstruction alone leaves underdetermined and that may underlie the cross-study variability of prior unsupervised PDAC taxonomies[2,3,14].

The central lesson is that survival supervision does not merely improve prediction; it resolves prognostically relevant biology into reproducible, source-attributable gene programs, the property that separates DeSurv from a supervised risk score and that should grow more valuable as clinically annotated cohorts accumulate. Together, these results support a model in which PDAC prognosis is encoded by coordinated malignant–stromal transcriptional states that outcome-supervised deconvolution can recover from bulk tumors.

Methods

Problem formulation and notation

Let $X \in \mathbb{R}_{\geq 0}^{p\times n}$ denote the nonnegative gene expression matrix. DeSurv approximates $X \approx WH$, where $W \in \mathbb{R}_{\geq 0}^{p\times k}$ contains nonnegative gene programs and $H \in \mathbb{R}_{\geq 0}^{k\times n}$ contains sample loadings. Additionally, let $y \in \mathbb{R}_{>0}^{n}$ denote patient survival times, $\delta \in \{0,1\}^n$ the censoring indicators ($\delta_i = 1$ if the event is observed), and $\beta \in \mathbb{R}^k$ the Cox regression coefficients. Survival outcomes are modeled through a Cox proportional hazards model with factor scores $Z = W^\top X \in \mathbb{R}^{k\times n}$ and linear predictor $Z^\top\beta$. Defining factor scores by projection onto the learned gene programs ($Z = W^\top X$) rather than by the inferred loadings H lets the same basis score new cohorts without refitting, the property we exploit for external validation.

The DeSurv Model

DeSurv integrates NMF with penalized Cox regression to identify gene programs associated with survival outcomes. The method operates in a survival-supervised framework: the reconstruction loss penalizes deviation from the observed expression data, while the survival term directs gene programs toward outcome-relevant structure. The parameter $\alpha \in [0,1)$ controls the relative contribution of each objective.

The joint objective is

$$\mathcal{L}(W,H,\beta) = (1-\alpha)\,\mathcal{L}_{\mathrm{NMF}}(W,H) - \alpha\,\mathcal{L}_{\mathrm{Cox}}(W,\beta), \tag{1}$$

where $\mathcal{L}_{\mathrm{NMF}}(W,H)$ is the NMF reconstruction error (including an ℓ_2 penalty on H) and $\mathcal{L}_{\mathrm{Cox}}(W,\beta)$ is the elastic-net penalized partial log-likelihood. A critical architectural choice is where survival supervision enters the factorization. Because factor scores are defined as $Z = W^\top X$, the Cox partial likelihood $\mathcal{L}_{\mathrm{Cox}}$ is a direct function of W and β, and the survival gradient acts explicitly on the gene program matrix W. The sample-level loadings H enter only through the reconstruction term and receive no survival gradient, preserving their interpretation as mixture coefficients[11,43]. When $\alpha = 0$, DeSurv reduces to standard unsupervised NMF. The Cox component can also accommodate additional sample-level covariates, such as tumor stage or grade, during optimization.

Optimization alternates updates for H, W, and β (SI Appendix, Section 2). The reconstruction term is minimized using multiplicative updates that preserve nonnegativity; the supervision term enters W's gradient as a Cox partial-likelihood derivative; and β is updated by Coxnet coordinate descent on the current factor scores $Z = W^\top X$ with elastic-net penalty parameters (λ, ξ). Although the joint problem is non-convex, the idealized alternating scheme converges to a stationary point under mild regularity conditions, and the implementation includes practical modifications (SI Appendix, Theorem 1). Closed-form update equations, the convergence proof, and BO control parameters are given in SI Appendix, Sections 1–3.

Hyperparameter selection and cross-validation

Hyperparameters $(k, \alpha, \lambda, \xi, n_{\text{top}})$ were selected by maximizing the cross-validated Harrell C-index using Bayesian optimization (BO)[44,45], which finds near-optimal configurations using far fewer objective evaluations than grid or random search over mixed integer–continuous hyperparameter spaces, with final rank k chosen by the one-standard-error rule (the smallest k whose predicted performance lies within one standard error of the maximum). All model selection was performed entirely within training data using nested cross-validation; no validation cohort data were used during any stage of tuning. Each fold was trained using multiple random initializations, and fold-level performance was defined as the average C-index across initializations. For stability, we used a consensus-based initialization for the final model, aggregating multiple DeSurv runs into a gene-gene co-occurrence matrix and constructing an initialization W_0 from the resulting clusters (SI Appendix, Section 6). Before projection, each learned gene program (column of W) was restricted to its BO-selected number of top genes (denoted $\widetilde{W}$) and column-normalized (ℓ_2-norm scaling) by the same procedure used during training (SI Appendix, Section 6); the resulting basis $\widetilde{W}$ was held fixed for all validation cohorts. In PDAC, BO selected $n_{\text{top}} = 270$ top genes per factor.

Because the goal is prognostic factorization, DeSurv selects k using the criterion that directly measures prognostic performance (cross-validated C-index) rather than the unsupervised diagnostics commonly used for NMF rank selection[12,24]. In the PDAC training data, standard NMF diagnostics (reconstruction residuals, cophenetic correlation, mean silhouette width) yielded inconsistent guidance across candidate ranks, a known limitation of unsupervised rank-selection criteria that can disagree on the same dataset (SI Appendix, Fig. S5). Cross-validated concordance varied across candidate ranks but plateaued from $k = 3$ through $k = 12$ (SI Appendix, Fig. S4). The one-standard-error rule applied to joint BO over rank and supervision strength accordingly selected the parsimonious $k = 3$ at $\alpha = 0.334$.

External validation cohorts were scored by projecting new datasets onto the learned programs via $Z = \widetilde{W}^{\top} X_{\text{new}}$, and survival associations were evaluated using Harrell's C-index, hazard ratios, and log-rank statistics; because Harrell's C-index can be optimistic under non-proportional hazards, we additionally report a proper score (the integrated Brier score and its index of prediction accuracy relative to a Kaplan–Meier reference) as a sensitivity check (SI Appendix, Proper-scoring sensitivity). Because the survival-supervised information lives in W rather than H, scoring is a pure matrix projection requiring no retraining or validation survival information. Methods that route survival supervision through H instead require sample-loading re-estimation on each new cohort[30,31]. Validation survival outcomes were therefore used only for downstream performance evaluation, not for model training, scoring, or hyperparameter tuning. Reported external hazard ratios and log-rank statistics therefore reflect purely out-of-sample generalization. Further training and validation details appear in SI Appendix, Part I.

Simulation studies

Simulation studies were conducted to assess recovery of prognostic latent structure and survival prediction under controlled conditions. Gene expression data were generated from a nonnegative factor model $X = WH$, where gene loadings W comprised three gene classes: marker genes, background genes, and noise genes. Marker genes were simulated to load strongly on a single factor and weakly on others, background genes to load strongly across all factors, and noise genes to have uniformly low loadings. Specifically, marker gene primary loadings were drawn from Gamma(shape $= 3$, rate $= 0.8$) and cross-loadings onto other factors from Gamma$(1, 20)$; background gene loadings from Gamma$(2, 1)$ across all factors; and noise gene loadings from Gamma$(1, 20)$. Sample-level factor activities H were drawn from Gamma$(2, 2)$. Each dataset comprised $G = 3{,}000$ genes, $N = 200$ samples, and $K = 3$ factors, with 150 marker genes per factor and 500 shared background genes. Cox regression coefficients were $\beta = (2, 0, 0)^\top$ in the prognostic scenario, $\beta = 0$ in the null scenario, and $\beta = (2, 0, 0)^\top$ in the mixed scenario where survival risk was driven by 300 genes comprising the 150 factor-1 markers and 150 randomly sampled background genes, creating partial overlap between the prognostic signal and variance-dominant outcome-neutral programs.

Survival times were generated from an exponential distribution in which risk depended on marker gene expression through $X^T W_{\text{true}}$, where W_{true} retained marker gene loadings for their corresponding factors and was zero otherwise; censoring times were generated independently from an exponential distribution. Each dataset was analyzed using DeSurv and standard NMF followed by Cox regression on inferred factors, both tuned using cross-validated concordance index via Bayesian optimization. Performance was summarized across repeated simulation replicates. We considered three simulation scenarios: a prognostic scenario in which prognostic programs explain low variance relative to outcome-neutral signals, a null scenario with no survival signal, and a mixed scenario where prognostic and variance-dominant programs partially overlap.

Real-world datasets

We analyzed publicly available RNA sequencing (RNA-seq) and microarray cohorts of PDAC with corresponding overall survival outcomes. Gene expression matrices were analyzed on a log scale: RNA-seq cohorts as $\log_2(\text{TPM} + 1)$; microarray cohorts in platform-normalized log-scale intensities. For each training cohort separately, genes were ranked by mean expression and by variance independently; the top 3,000 genes with the lowest sum of these two ranks were retained, selecting genes that are both highly expressed and highly variable; the intersection across training cohorts yielded 1,970 genes used for all analyses. Validation datasets were restricted to this same gene set. Survival times and censoring indicators were taken from the associated clinical annotations. Of the seven PDAC cohorts we considered, two were used for training (TCGA and CPTAC) and five were used for external validation (Dijk, Moffitt, PACA-AU array, PACA-AU RNA-seq, and Puleo). To harmonize differences in scale across cohorts, filtered gene expression data were within-subject rank transformed before model training (ranks are inherently nonnegative, preserving NMF compatibility); ranking removes platform- and library-size differences in expression scale that would otherwise dominate cross-cohort projection. More details about the datasets can be found in SI Appendix, Section 7.

Per-cohort participant flow is summarized in SI Appendix, Table S1. Of 321 pooled training-cohort samples (TCGA-PAAD, $n = 181$; CPTAC-3, $n = 140$), 48 were excluded for missing survival time, missing event indicator, or quality-control failure, yielding 273 analytic training samples (139 events). Of 979 pooled validation-cohort samples, 363 were excluded for non-PDAC histology, non-tumor tissue, or missing survival annotation, yielding 616 analytic validation samples (414 events) across five cohorts: Dijk ($n = 90$, 81 events), Moffitt GEO array ($n = 123$, 83), PACA-AU array ($n = 63$, 38), PACA-AU RNA-seq ($n = 52$, 31), and Puleo array ($n = 288$, 181).

Data Availability

The training and validation datasets used in this study are publicly available and de-identified; no Institutional Review Board (IRB) approval was required for their analysis. Training data were obtained from the Genomic Data Commons: TCGA-PAAD[46] and CPTAC-3[47]. Validation cohorts were obtained from ArrayExpress (Dijk, E-MTAB-6830[48]; Puleo, E-MTAB-6134[49]), the Gene Expression Omnibus (GEO; Moffitt, GSE71729[50]), and the International Cancer Genome Consortium (ICGC) Data Portal (PACA-AU array and RNA-seq; European Genome–phenome Archive (EGA) study EGAS00001000154[51]). Expression data and molecular classifications (PurIST, DeCAF) for the validation cohorts were originally compiled for the DeCAF study[5]; cohort details and sample sizes after quality control are summarized in SI Appendix, Section 7. The treated bulk-tissue cohorts used in the exploratory translational analysis (RASH-ACCEPT and the Linehan FOLFIRINOX$_{\pm}$CCR2-inhibitor series) are restricted-access clinical-trial datasets that are not publicly deposited; raw data are available from the respective study investigators under a data-use agreement, subject to the governing trial consents and institutional approvals. To preserve reproducibility without redistributing restricted data, the derived per-cohort summary statistics reported in the main text and SI Appendix are provided in the reproducibility repository (file `results/treated_cohort_stats.rds`), so that every treated-cohort value can be inspected and verified.

Code Availability

All analyses were performed in R (version 4.6.0; R Core Team) using DeSurv (v1.0.1; this work), `NMF`[43], `glmnet`[52], `survival`, `ggplot2`, and `cowplot`; Bayesian optimization for hyper-parameter selection was implemented within DeSurv. The DeSurv R package is available at https://github.com/rashidlab/DeSurv (archived at Zenodo, DOI `10.5281/zenodo.20150929`). Reproducibility code, processed data, and the exact session-info file are available at https://github.com/rashidlab/DeSurv-paper (archived at Zenodo, DOI `10.5281/zenodo.20150946`); this includes `code/11_treated_cohort_analysis.R`, which generates the treated-cohort summary statistics above.

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Inclusion and Diversity. The patient cohorts analyzed in this study were curated by the original investigators or consortia (TCGA, CPTAC, ICGC, and the original Moffitt, Dijk, and Puleo studies) and our access was limited to the de-identified expression and survival data they provide; we did not stratify analyses by sex, race, or ethnicity because consistent demographic annotations were not available across all cohorts. Sex and other demographic variables that could modify the prognostic relevance of the gene programs identified here remain important directions for follow-up work as more comprehensively annotated cohorts become available.

Author Contributions

Conceptualization: A.M.Y., N.U.R. Methodology: A.M.Y., A.Y., D.L., N.U.R. Software: A.M.Y. Formal analysis: A.M.Y. Investigation: A.M.Y. Data curation: A.M.Y., X.L.P. Visualization: A.M.Y. Writing – original draft: A.M.Y., N.U.R. Writing – review and editing: A.M.Y., A.Y., X.L.P., J.J.Y., D.L., N.U.R. Resources (PDAC domain expertise and biological interpretation): X.L.P., J.J.Y. Supervision: N.U.R. Funding acquisition: A.M.Y., N.U.R. All authors reviewed and approved the final manuscript.

Competing Interests

N.U.R. and J.J.Y. are named as inventors on US patents US20220127676A1 and US20250066856A1, which relate to pancreatic cancer subtype classification. The PurIST and DeCAF classifiers were used only as published, independently derived comparators: their subtype calls were obtained with the released classifiers and included as fixed covariate annotations, and the adjustment analyses using these calls are objective and reproducible from the deposited code. They are not components of DeSurv. The other authors declare no competing interests.

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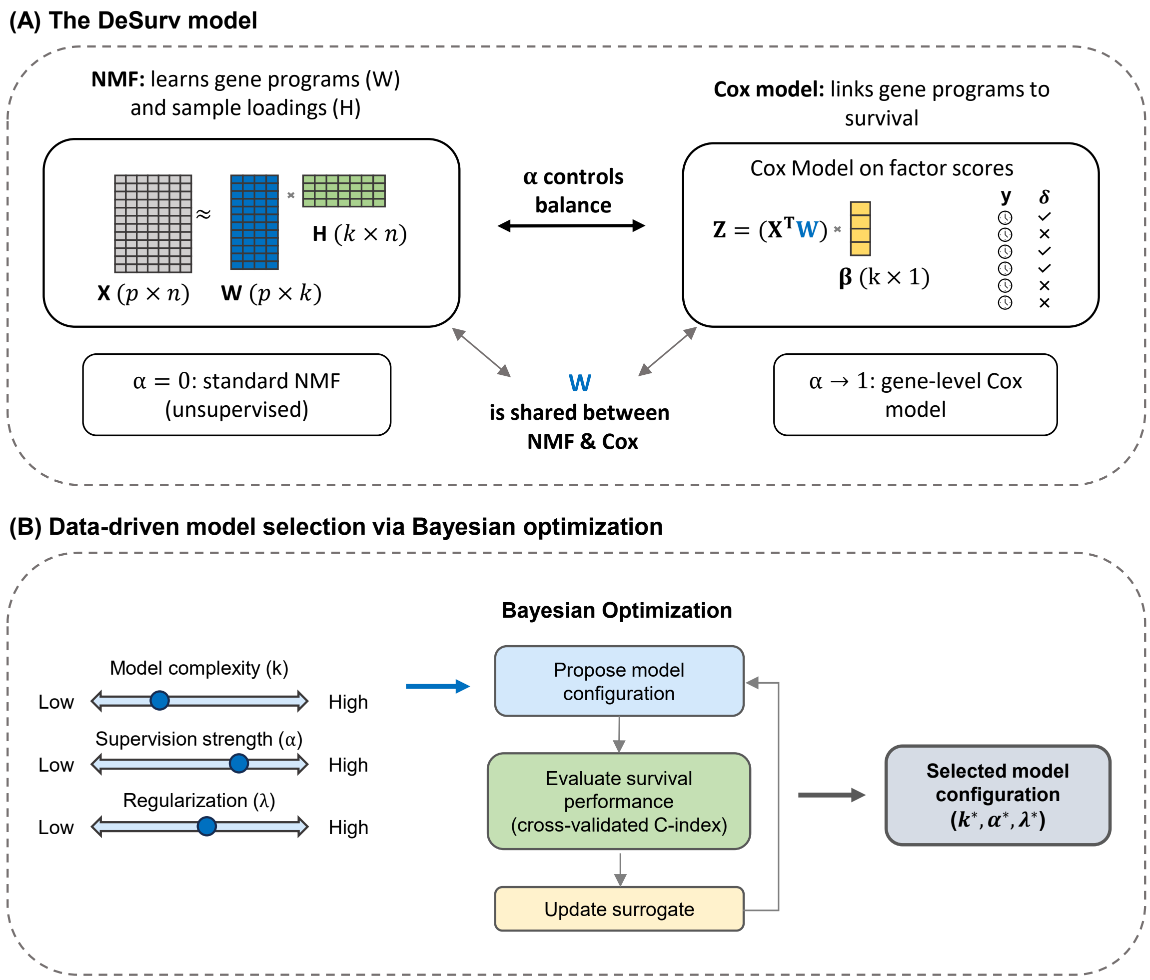

Figure 1: Overview of the DeSurv framework. (A) DeSurv jointly optimizes NMF reconstruction ($X \approx WH$) and Cox proportional hazards likelihood. The gene program matrix W is shared between objectives: factor scores $Z = W^\top X$ serve as covariates in the Cox model with coefficients β. A supervision parameter $\alpha \in [0, 1)$ controls the balance between reconstruction fidelity ($\alpha = 0$, standard NMF) and survival prediction (α close to 1). (B) The factorization rank (k), supervision strength (α), and regularization (λ) are selected via Bayesian optimization over cross-validated concordance index.

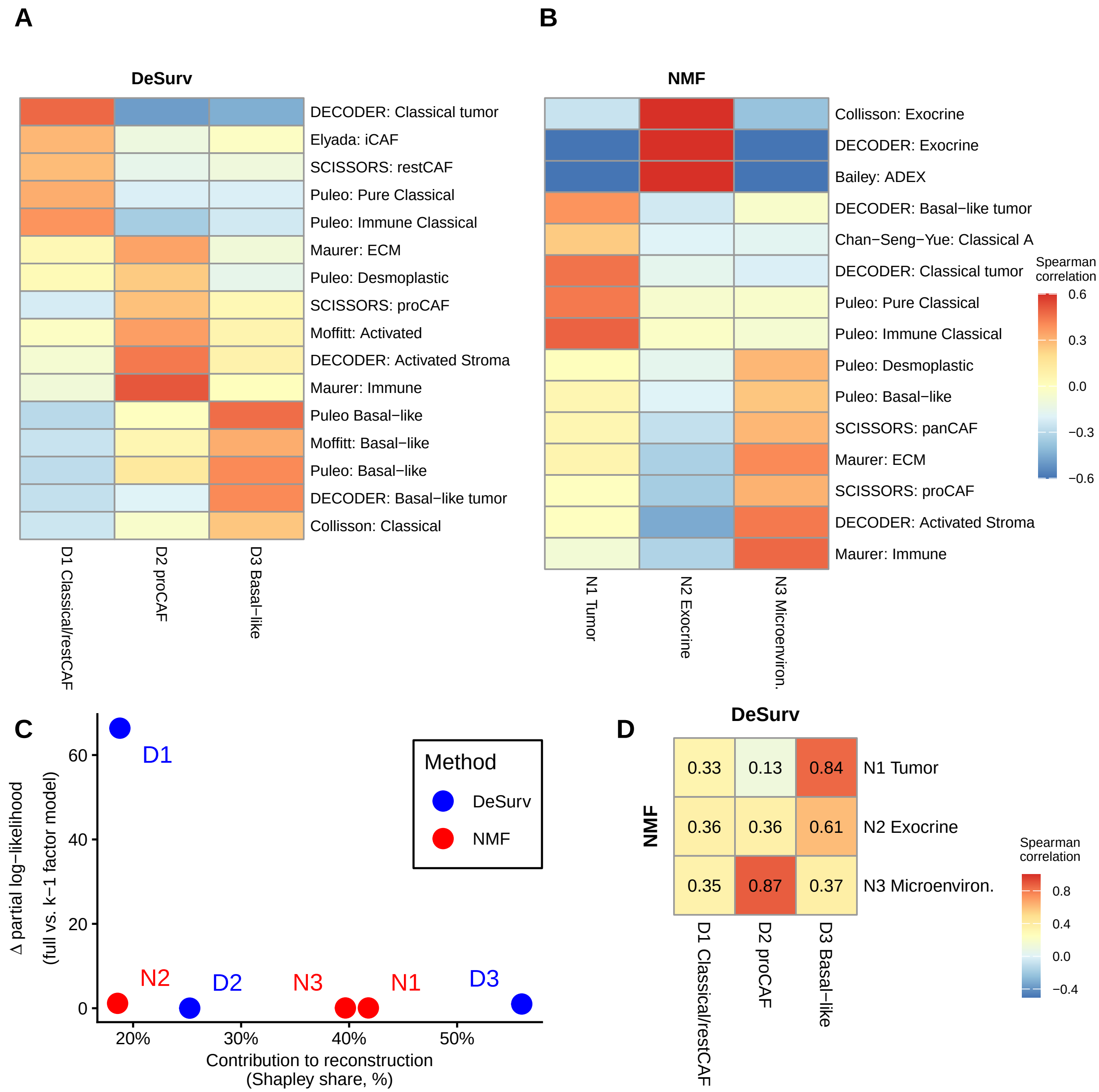

Figure 2: Survival supervision produces different factor structures in PDAC (TCGA + CPTAC training cohort, $n = 273$ patients, 139 events). (A–B) Spearman correlations between factor gene rankings (top 50 genes per factor) and established PDAC gene programs for DeSurv (A) and standard NMF (B) at $k = 3$. Only $|r| > 0.2$ shown. DeSurv concentrates survival signal into D1, whose Classical-tumor + restCAF-like-stroma pole marks lower risk, while suppressing exocrine variation; standard NMF devotes one factor to exocrine signals. (C) Per-factor contribution to reconstruction of the expression matrix (normalized reconstruction Shapley share, which partitions and sums to 100%) versus survival contribution (Δ Cox partial log-likelihood upon factor removal). DeSurv concentrates survival signal into D1 while NMF's largest-reconstruction factors — including the exocrine-associated factor N2 — contribute negligibly to survival. (D) Pairwise correspondence between NMF and DeSurv factors, measured by Spearman rank correlation of W-matrix gene loadings over all genes.

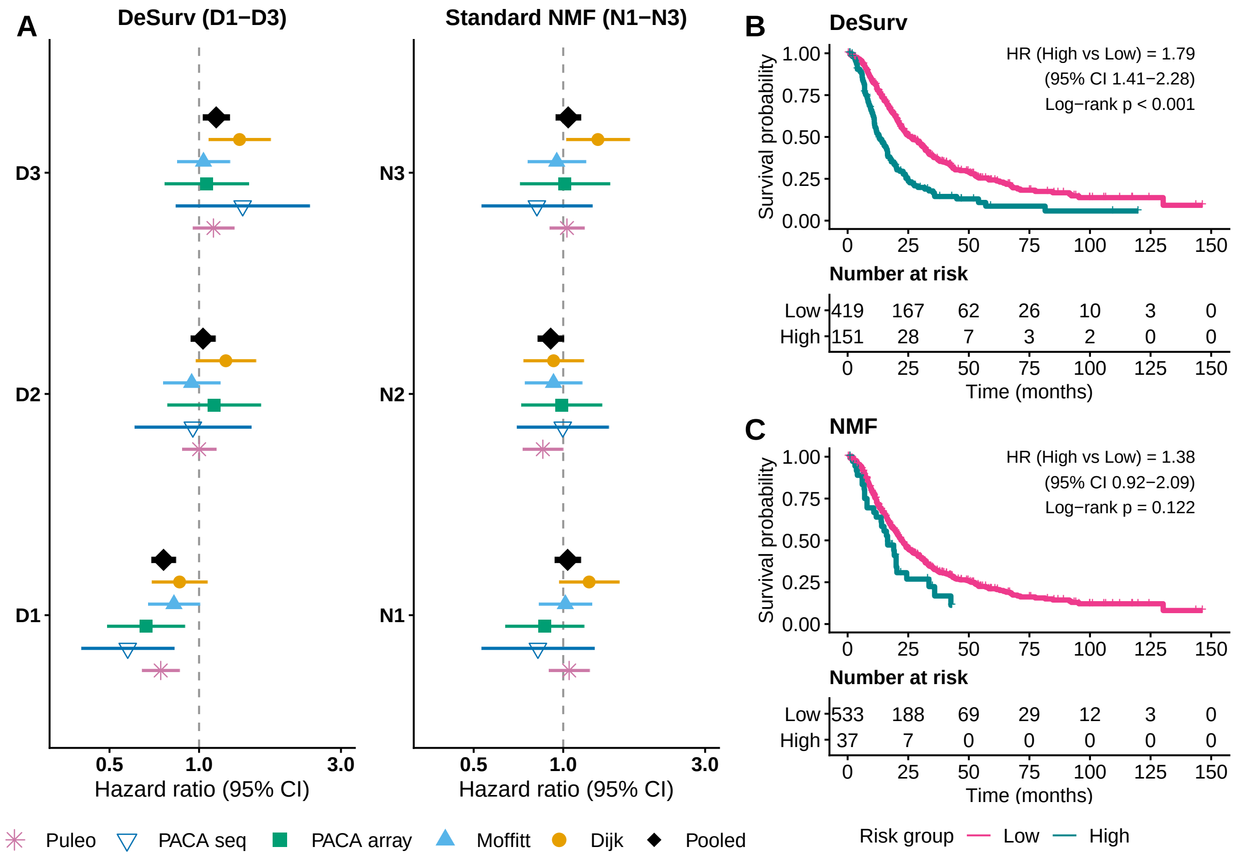

Figure 3: Survival-aligned programs generalize across independent PDAC cohorts (pooled $n = 616$ patients across 5 cohorts: Dijk $n = 90$; Moffitt GEO array $n = 123$; PACA-AU array $n = 63$; PACA-AU RNA-seq $n = 52$; Puleo array $n = 288$). (A) Forest plot of univariate hazard ratios (95% CIs) from separate Cox models for each factor, showing each factor's marginal survival association in each validation cohort. DeSurv factor D1 shows a consistent protective association across all five cohorts (pooled HR = 0.76), while no standard NMF factor achieves a comparable pooled effect. Error bars are 95% confidence intervals; black diamonds denote inverse-variance-weighted pooled estimates. (B–C) Kaplan–Meier curves for pooled external validation samples stratified by the combined multivariate linear predictor ($\hat{\beta}_1 Z_1 + \hat{\beta}_2 Z_2 + \hat{\beta}_3 Z_3$, where coefficients are learned from training data), dichotomized at a cross-validated optimal cutpoint: DeSurv (B) and standard NMF (C). No validation-cohort survival outcomes were used in model training, scoring, or hyperparameter selection.

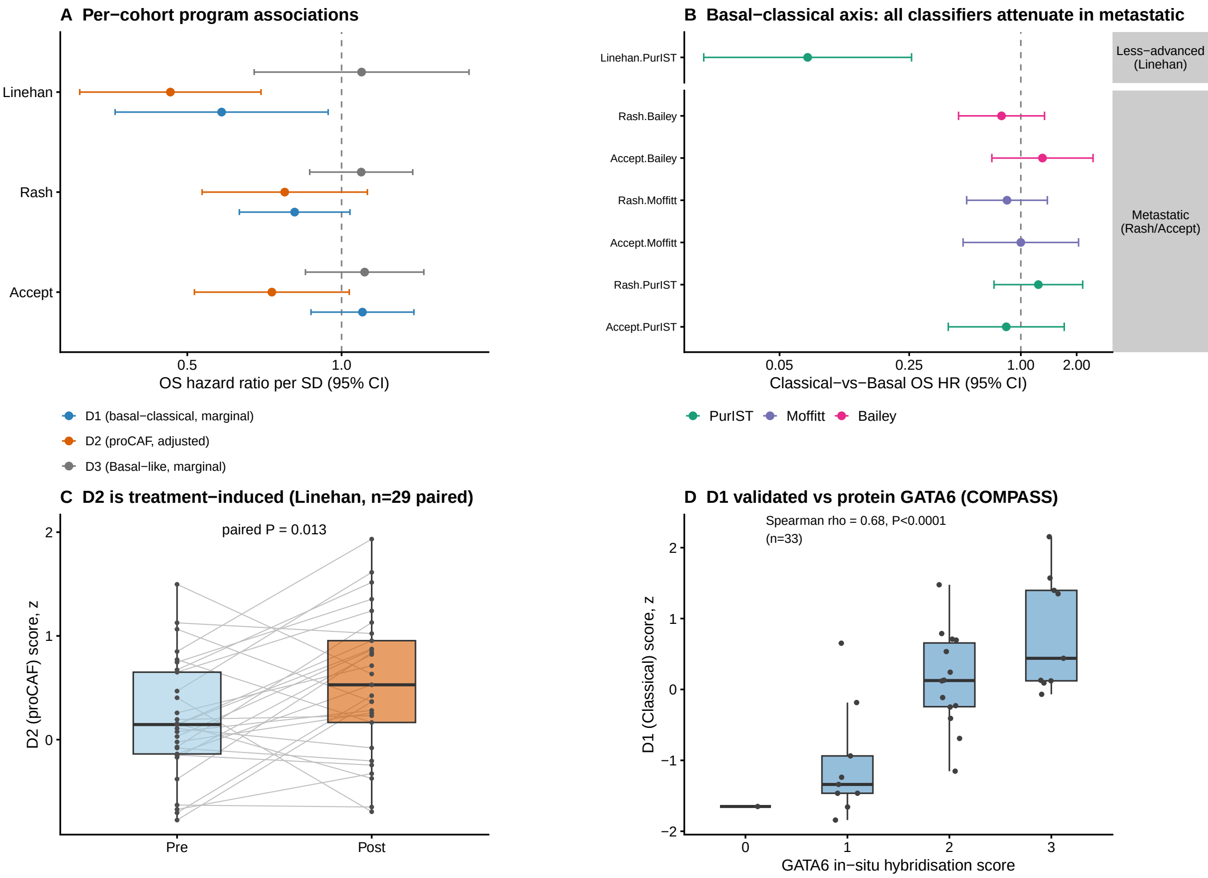

Figure 4: Survival-supervised programs show context-dependent prognostic associations in treated PDAC. Trained DeSurv programs were projected without retraining onto three bulk treated cohorts (Linehan, borderline-resectable/locally-advanced, $n = 66$; Rash, metastatic, $n = 85$; Accept, metastatic, $n = 70$) and analyzed per cohort. (A) Per-cohort overall-survival hazard ratios per SD for each program (D1 and D3 marginal; D2 PurIST + DeCAF-adjusted); D2 (proCAF) is protective in all three cohorts, D1 (basal–classical) is prognostic only in the less-advanced cohort, and D3 is null on overall survival. (B) The basal–classical prognostic axis is setting-dependent and concordant across measures: Classical-vs-Basal overall-survival hazard ratios for the PurIST, Moffitt, and Bailey classifiers in less-advanced versus metastatic disease, all attenuating together in metastatic disease. (C) The proCAF program D2 increases from pre- to post-treatment in paired biopsies (paired Wilcoxon test). (D) Orthogonal validation: the D1 score versus GATA6 RNA in situ hybridization (Spearman correlation).

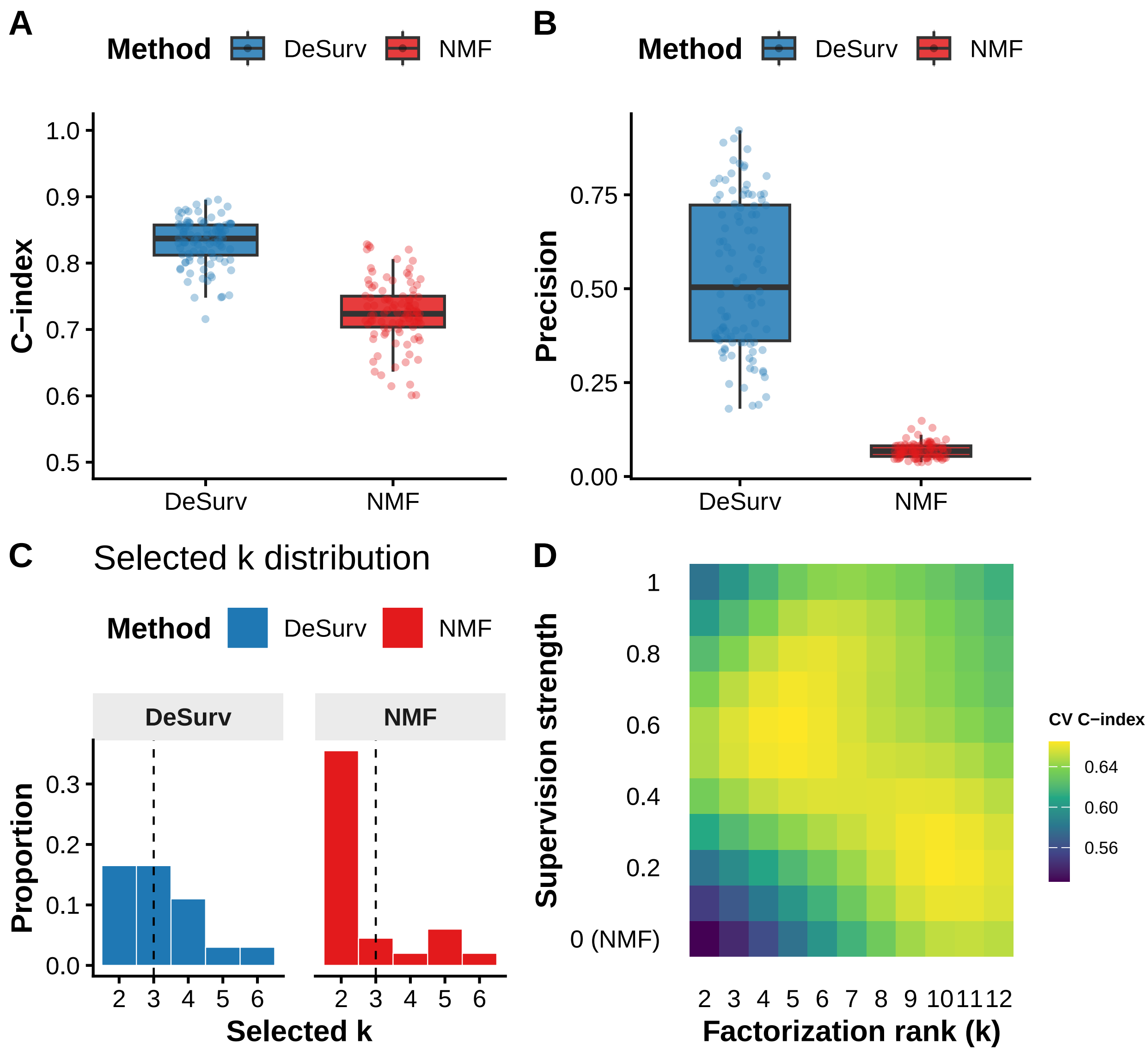

Figure 5: Survival supervision improves recovery of true prognostic gene programs in controlled simulations. (A) Test-set concordance index across 100 simulation replicates (p = 3,000 genes, n = 200 samples, true k = 3) comparing DeSurv to standard NMF ($\alpha = 0$); paired Wilcoxon $P < 0.001$. (B) Precision (fraction of a factor's top-weighted genes, defined in main text, overlapping a true prognostic program) across replicates. In (A) and (B), boxes show median and interquartile range (IQR); whiskers extend to 1.5×IQR; points beyond are individual replicates. The y-axis in (A) is truncated at 0.5 because C-index values below 0.5 indicate worse-than-random concordance. (C) Distribution of selected k values across the same 100 replicates for DeSurv versus standard NMF. (D) Gaussian-process–predicted mean cross-validated C-index from Bayesian optimization over factorization rank (k) and supervision strength (α) in PDAC training data (TCGA + CPTAC), showing the broad plateau spanning k = 3–12 from which the one-standard-error rule selects $k = 3$ (Methods).